

Data analytics capstone project

Exploratory Data Analysis of Netflix content



Aim

The primary aim of this project is to perform Exploratory Data Analysis (EDA) on the Netflix dataset to gain meaningful insights into the platform's content library. This project seeks to understand the structure, distribution, trends, and patterns of movies and TV shows available on Netflix by analyzing various attributes such as content type, genres, release years, duration, ratings, countries, and languages. Through systematic data cleaning, transformation, and visualization, the project aims to identify how Netflix's content has evolved over time, how it is distributed across different regions, and how different genres and categories contribute to content diversity. The analysis also focuses on uncovering relationships between variables—such as ratings and duration—to better understand content characteristics. Additionally, this project aims to explore user preference indicators, such as the popularity of specific genres or content types, and to evaluate Netflix's strategy in terms of content variety, geographical reach, and audience targeting. By leveraging visual analytics using tools like Plotly, the project transforms raw data into clear, interpretable insights that support data-driven conclusions and recommendations.

Ultimately, the goal of this EDA project is to enhance analytical thinking, strengthen data visualization skills, and demonstrate how exploratory analysis can be used to extract actionable insights from real-world datasets.

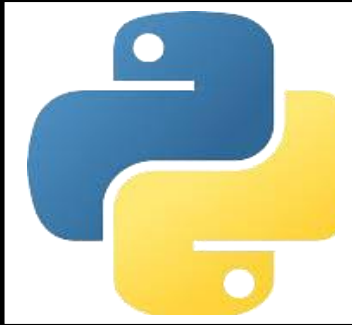
Objective

The objective of this Exploratory Data Analysis (EDA) project is to analyze and understand the Netflix content dataset in order to uncover meaningful insights and trends. The project involves exploring the structure of the dataset, cleaning and preprocessing the data by handling missing values and correcting inconsistencies, and performing descriptive statistical analysis. Various visualizations are used to examine the distribution of content across genres, content types, countries, languages, and ratings. Time-based analysis is conducted to identify trends in content addition and to study how characteristics such as duration and maturity ratings have evolved over the years. The project also evaluates content diversity, investigates potential relationships between variables such as ratings and duration, and infers content popularity based on genre and content type distribution. Where direct user engagement data is unavailable, sentiment analysis of content descriptions is performed as a proxy. Overall, the project aims to provide a comprehensive understanding of Netflix's content strategy and global reach through data-driven insights and interactive visualizations.

Purpose

The purpose of this Netflix Exploratory Data Analysis (EDA) project is to explore and analyze Netflix's content dataset to uncover meaningful insights about the platform's content library. The project focuses on understanding the distribution and diversity of movies and TV shows across genres, ratings, languages, countries, and content types, as well as examining how these characteristics have evolved over time. By performing data cleaning, descriptive analysis, and visual exploration, the study aims to identify trends, patterns, and relationships between key variables such as duration, ratings, and genres. In the absence of direct user engagement data, content volume is used to infer popularity and user preferences. Overall, this analysis helps provide a deeper understanding of Netflix's content strategy, global reach, and audience-focused offerings through data-driven insights and visualizations.

Tools



Python

Use: A high-level programming language widely used for data analysis, automation, and AI.

Features:

Easy to learn and read.

Huge library support for data, web, and AI.

Backbone for libraries like Pandas, NumPy, Matplotlib, and



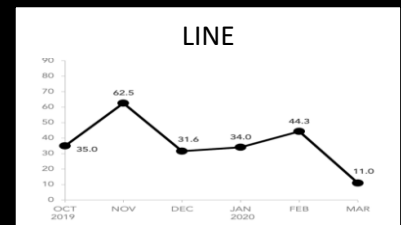
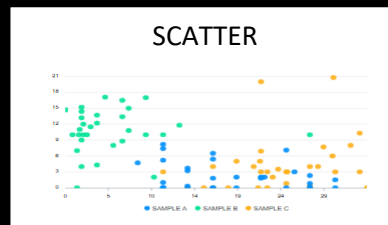
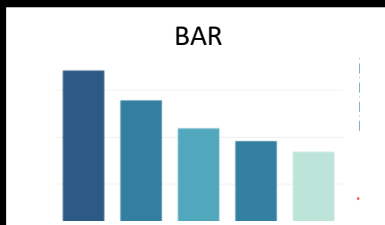
Plotly

Plotly is an interactive data visualization library used to create dynamic and engaging charts.

It supports various visualizations such as bar charts, line charts, scatter plots, pie charts, and maps.

Plotly allows users to interact with data through features like zooming, hovering, and filtering.

Graphs



problem statements

TASKS OVERVIEW:-

TASK 1 :- BEGIN BY LOADING AND EXPLORING THE DATASET.

TASK 2 :- CLEAN AND PREPROCESS DATA TO ENSURE CONSISTENCY AND ACCURACY FOR ANALYSIS.

TASK 3 :- COMPUTE BASIC DESCRIPTIVE STATISTICS SUCH AS MEAN, MEDIAN, MODE, RANGE, AND STANDARD DEVIATION FOR RELEVANT

TASK 4 :- CREATE VISUALIZATIONS TO REPRESENT THE DISTRIBUTION OF CONTENT OVER DIFFERENT GENRES.

TASK 5 :- VISUALIZE THE DISTRIBUTION OF CONTENT ACROSS RELEASE YEARS.

TASK 6 :- EXPLORE THE GEOGRAPHICAL DISTRIBUTION OF CONTENT (IF APPLICABLE).

TASK 7 :- IF THERE'S A TEMPORAL COMPONENT, PERFORM TIME SERIES ANALYSIS TO IDENTIFY TRENDS AND PATTERNS OVERTIME.

TASK 8 :- ANALYZE THE DISTRIBUTION OF CONTENT RATINGS.

TASK 9 :- EXPLORE THE LENGTH OF MOVIES OR EPISODES AND IDENTIFY ANY TRENDS.

TASK 10 :- IDENTIFY AND PRESENT TOP-RATED MOVIES OR TV SHOWS BASED ON USER RATINGS.

TASK 11 :- ANALYZE TRENDS IN THE POPULARITY OF DIFFERENT GENRES OVER TIME.

TASK 12 :- FURTHER EXPLORE THE DISTRIBUTION OF CONTENT ACROSS DIFFERENT COUNTRIES AND REGIONS.

TASK 13 :- INVESTIGATE POTENTIAL CORRELATIONS BETWEEN VARIABLES (E.G., RATINGS AND DURATION).

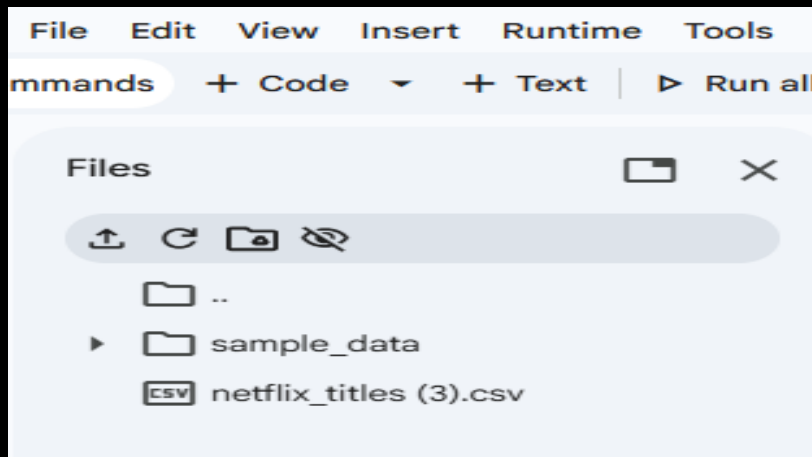
TASK 14 :- EVALUATE THE DIVERSITY OF CONTENT BY ANALYZING THE NUMBER OF UNIQUE GENRES AND CATEGORIES.

TASK 15 :- EXPLORE HOW THE CHARACTERISTICS OF CONTENT (E.G., DURATION, RATINGS) HAVE EVOLVED OVER THE YEARS.

OVERVIEW

IMPORT DATASETS RELATED TO NETFLIX CONTENT AND STATE CODES.

DATA IMPORTED: -



OVERVIEW :-

Here we imported/ load the CSV / Excel file on which we are going to analysis the data.

IMPORT PYTHON LIBRARIES:-

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Overview :-

To perform analysis we have to import few libraries of python i.e: Pandas (for data cleaning and manipulatio), Matplotlib and seaborn (for Visualisation), & Numpy (for performing numerical operations).

Read the CSV / Excel file

```
df = pd.read_csv("/content/netflix_titles (3).csv")
```

df								
	show_id	type	title	director	cast	country	date_added	
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mablane, Thaban...	South Africa	September 24, 2021	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	

8807 rows x 12 columns

OVERVIEW :-

Using pandas library ,we read the first excel file . Which contains 8807 rows and 12 columns.

Check data shape & describe the numerical values of dataset:

```
df.shape
```

```
(8807, 12)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 8807 entries, 0 to 8806  
Data columns (total 12 columns):  
#   Column                Non-Null Count  Dtype    
---  -  
0   show_id                8807 non-null   object   
1   type                  8807 non-null   object   
2   title                 8807 non-null   object   
3   director              6173 non-null   object   
4   cast                  7982 non-null   object   
5   country               7976 non-null   object   
6   date_added            8797 non-null   object   
7   release_year          8807 non-null   int64    
8   rating                8803 non-null   object   
9   duration              8804 non-null   object   
10  listed_in             8807 non-null   object   
11  description            8807 non-null   object   
dtypes: int64(1), object(11)
```

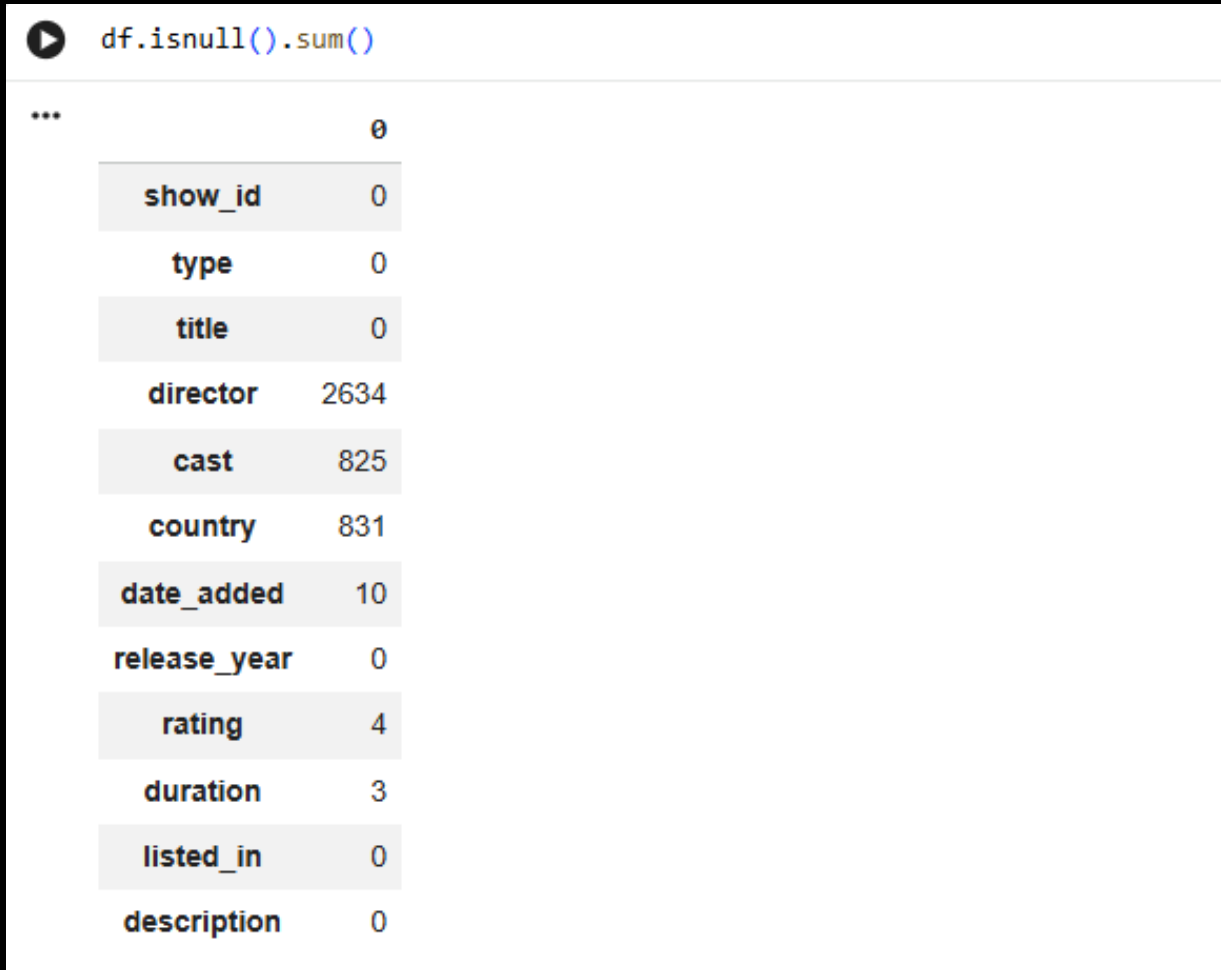
```
df.describe()
```

	release_year
count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

OVERVIEW:-

This Dataframe is confined to have 8807 rows and 12 column and using `df.describe()` we found that there is only one numerical column (`release_year`) .

Null values



A screenshot of a Jupyter Notebook cell. The code `df.isnull().sum()` is entered in the input area. The output is a series of 12 columns and their corresponding null counts. The columns are: `show_id` (0), `type` (0), `title` (0), `director` (2634), `cast` (825), `country` (831), `date_added` (10), `release_year` (0), `rating` (4), `duration` (3), `listed_in` (0), and `description` (0). The output is displayed in a table-like format with alternating row colors.

...	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

OVERVIEW:-

As we can see that this dataframe consists of 12 columns and by `df.isnull()` we find that there are 6 columns which hold null \Nan values .

Filling up null values

```
df.director.fillna ("unknown", inplace =True)
```

```
df.country.fillna ("unknown", inplace =True)
```

```
df.cast.fillna ("unknown", inplace =True)
```

```
df.loc[5989, "rating"] = "TV-PG"  
df.loc[6827, "rating"] = "TV-14"  
df.loc[7312, "rating"] = "TV-PG"  
df.loc[7537, "rating"] = "PG-13"
```

```
df.loc[6066, "date_added"] = "october 2,2013"  
df.loc[6174, "date_added"] = "April 14, 2013"  
df.loc[6795, "date_added"] = "january 15, 1993"  
df.loc[6806, "date_added"] = "September 22, 1994"  
df.loc[6901, "date_added"] = "july 23, 2003"  
df.loc[7196, "date_added"] = "august 14, 2004"  
df.loc[7254, "date_added"] = "october 5, 2002"  
df.loc[7406, "date_added"] = "May 3, 2013"  
df.loc[7847, "date_added"] = "june 18, 2003"  
df.loc[8182, "date_added"] = "may 13, 2012"
```

OVERVIEW:-

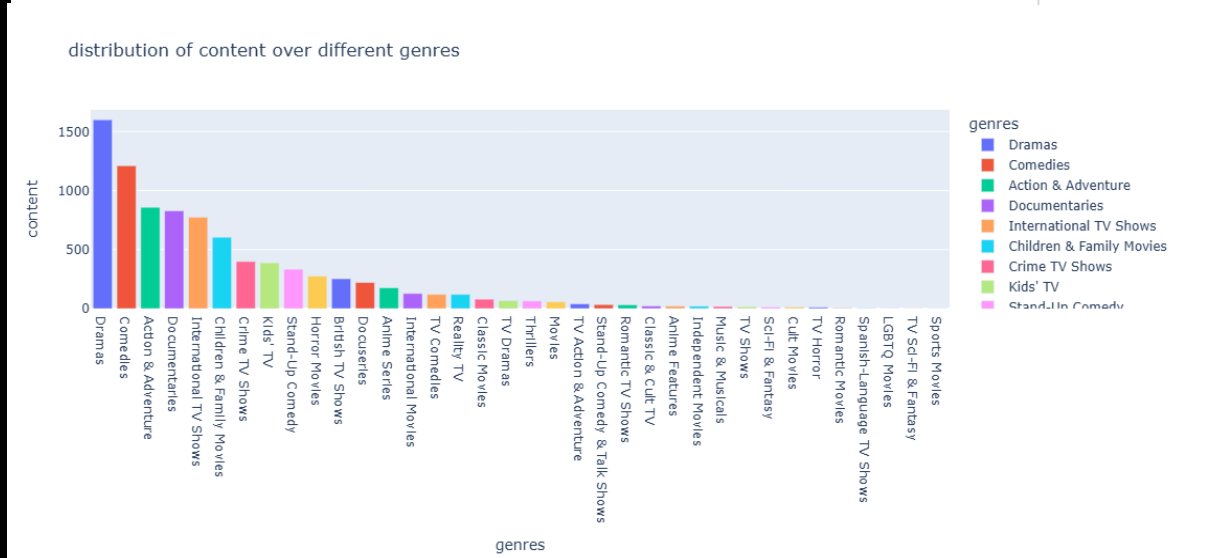
Our dataframe consists of 6 null columns . So we fill up columns like (director, country, cast) by global variable “unknown” and columns like (date_added and rating) by actual data values.

Insights

DISTRIBUTION OF CONTENT OVER DIFFERENT GENRES.

TASK:-4 CREATE VISUALIZATIONS TO REPRESENT THE DISTRIBUTION OF CONTENT OVER DIFFERENT GENRES.

```
import plotly.express as px
px.bar(a,
      x = "listed_in",
      y = "count",
      color = "listed_in", labels = {"listed_in": "genres",
      "count": "content"},
      title = "distribution of content over different genres")
```



OVERVIEW:-

This Plotly bar chart illustrates the distribution of content (movies and TV shows) across different genres. The x-axis represents genres (listed_in), while the y-axis shows the count of content titles in each genre. Color coding by genre helps visually differentiate categories and compare their relative volumes.

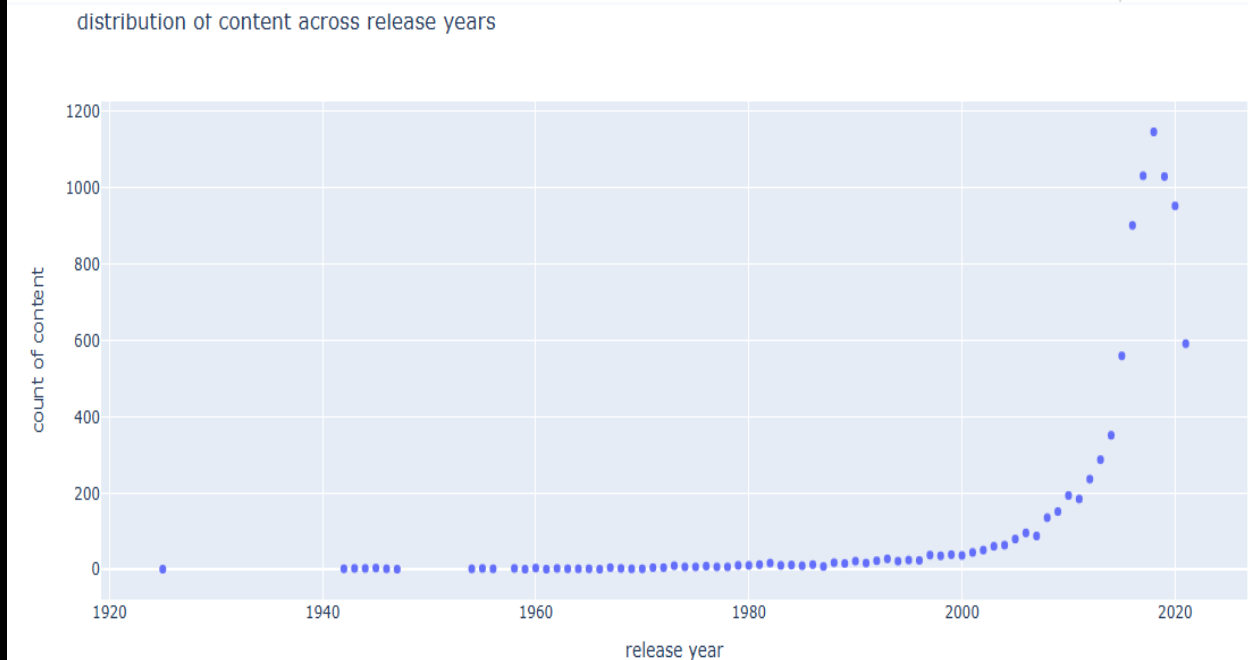
OUTCOME:-

Drama has the highest content count, making it the most dominant genre on the platform. Comedies, Action & Adventure, and Documentaries also contribute a large share, indicating strong audience demand for these genres.

TV-focused genres such as Kids' TV, International TV Shows, and Stand-Up Comedy have noticeable representation, reflecting growing episodic content consumption.

DISTRIBUTION OF CONTENT ACROSS RELEASE YEARS

```
# TASK 5 :- Visualize the distribution of content across release years.  
import plotly.express as px  
px.scatter(b, x='release_year', y="count",  
          labels = {"release_year": "release year", "count":  
                  "count of content"}, title = " distribution of content across release years")
```



OVERVIEW:-

This Plotly scatter plot represents the distribution of content (movies and TV shows) across different release years. The x-axis shows the release year, while the y-axis indicates the number of content titles released in each year, helping visualize historical trends over time.

OUTCOME :-

- a) Very low content production before the 1980s, indicating limited availability of older titles.
- b) A steady rise from the 1990s, showing gradual growth in content creation
- c) A sharp spike after 2010, reflecting rapid expansion of streaming platforms and digital content.
- D) Recent years (2018–2020) record the highest content counts, highlighting peak production activity.

GEOGRAPHICAL DISTRIBUTION OF CONTENT

```
# task 6 :- Explore the geographical distribution of content (if applicable).  
c= px.choropleth(c,locations= c["country"],locationmode="country names",  
                projection = "orthographic",color = c['count'],title=  
                " geographical distribution of content")
```

geographical distribution of content



OVERVIEW:-

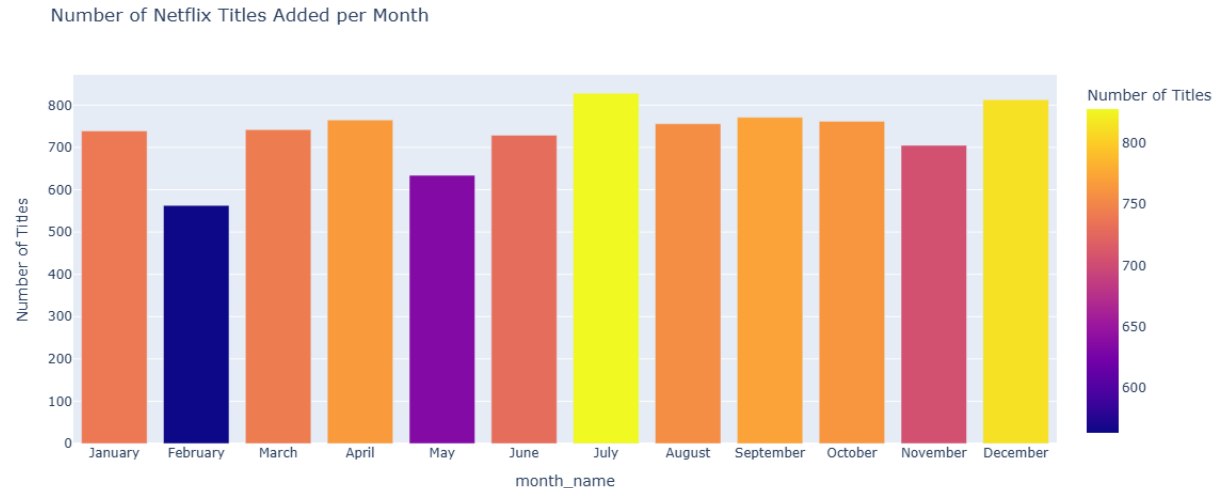
This Plotly choropleth world map visualizes the geographical distribution of content (movies and TV shows) across different countries. Each country is shaded based on the number of content titles produced or available, with darker/brighter colors representing higher counts.

OUTCOME:-

- a) The United States dominates the content landscape, contributing the highest number of titles.
- B) North America and parts of Europe show strong content representation, indicating mature entertainment industries.
- C) Emerging markets in Asia, South America, and Africa have comparatively lower content counts but show global reach.
- D) The distribution highlights a concentration of content production in a few key countries, with gradual global diversification.

NUMBER OF NETFLIX TITLES ADDED PER MONTH

```
# TASK 7 :- IF THERE'S A TEMPORAL COMPONENT, PERFORM TIME SERIES ANALYSIS TO IDENTIFY TRENDS AND PATTERNS OVERTIME.  
import plotly.express as px  
px.bar(d,  
      x='month_name',  
      y='count',  
      title='Number of Netflix Titles Added per Month',  
      labels={'month': 'Months', 'count': 'Number of Titles'},color = "count")
```



OVERVIEW:-

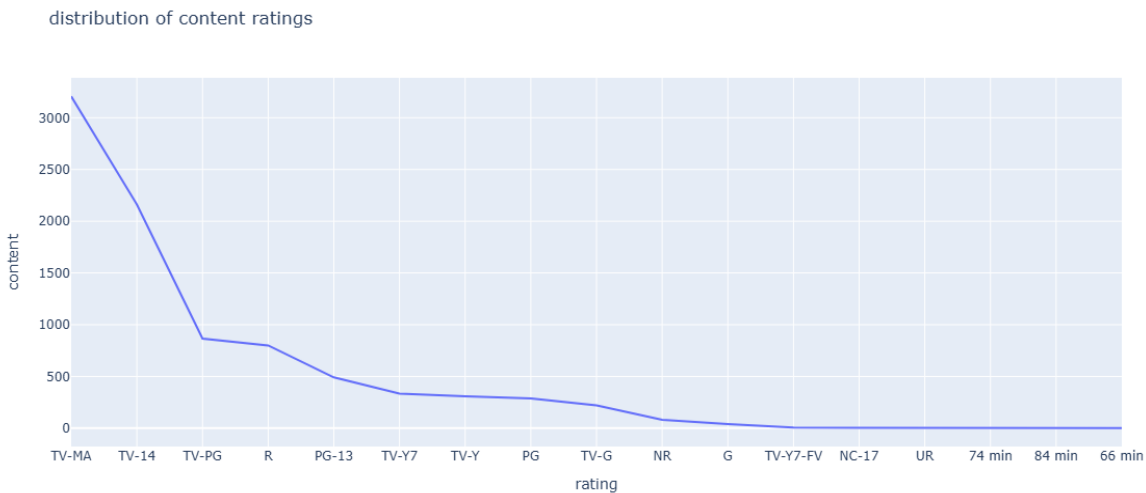
This Plotly bar chart shows the number of Netflix titles added per month. The x-axis represents the months of the year, while the y-axis indicates the count of titles added. Color intensity highlights variations in monthly additions, making seasonal trends easy to spot.

OUTCOME:-

- July and December have the highest number of content additions, indicating peak release periods.
- February shows the lowest number of titles added, suggesting slower content onboarding early in the year.
- Content additions remain fairly consistent from March to October, reflecting a steady release strategy.
- The year-end spike points to strategic releases during holiday seasons to maximize viewer engagement.

DISTRIBUTION OF CONTENT RATINGS

```
# TASK 8 :- ANALYZE THE DISTRIBUTION OF CONTENT RATINGS.  
import plotly.express as px  
px.line(e, x= "rating",y = "count",  
labels = {"count":"content"},  
title = " distribution of content ratings")
```



OVERVIEW:-

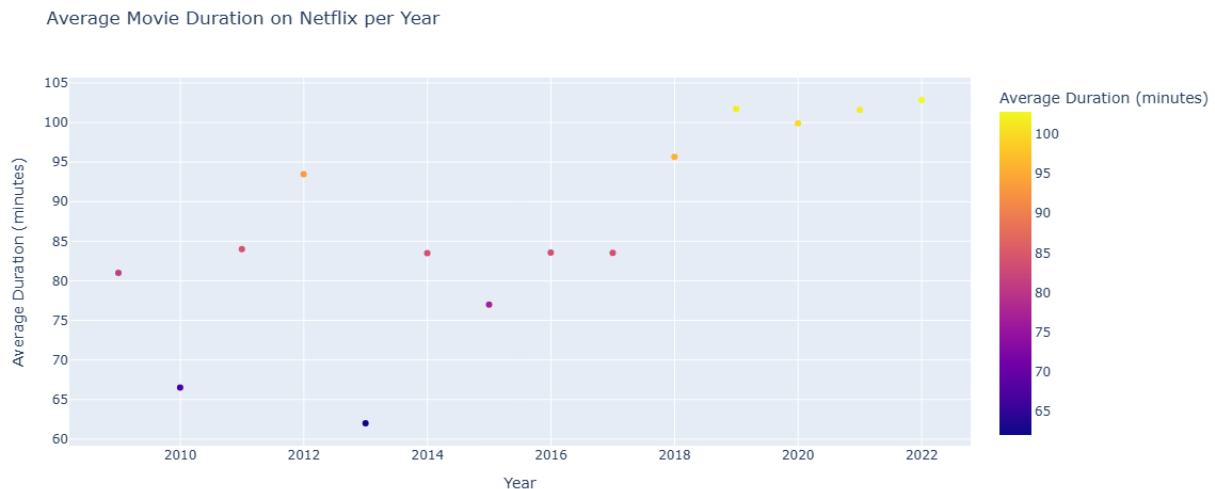
This Plotly line chart shows the distribution of content ratings across the platform. The x-axis represents different content ratings (e.g., TV-MA, TV-14, PG-13), while the y-axis shows the count of content titles under each rating, highlighting how content is classified by audience suitability.

OUTCOME:-

- A) TV-MA has the highest content count, indicating a strong focus on mature-audience content.
- b) TV-14 is the second most common rating, showing a significant volume of teen-friendly content.
- c) Ratings such as PG-13, R, and TV-PG have moderate representation, balancing family and adult audiences.
- d) Kids-focused ratings (TV-Y, TV-Y7, TV-G) have comparatively fewer titles.
- e) Rare ratings (NC-17, UR, NR) contribute negligible content, indicating limited use.

Average Movie Duration on Netflix per Year

```
# TASK 9 :- EXPLORE THE LENGTH OF MOVIES OR EPISODES AND IDENTIFY ANY TRENDS.  
import plotly.express as px  
px.scatter(f, x="date_added", y="duration_min",  
          title="Average Movie Duration on Netflix per Year",  
          labels={"date_added": "Year",  
                  "duration_min": "Average Duration (minutes)" },color = "duration_min")
```



OVERVIEW:-

This Plotly scatter plot shows the average movie duration on Netflix per year. The x-axis represents the year, while the y-axis indicates the average duration (in minutes) of movies released or added in that year. Color intensity reflects variations in average runtime across years.

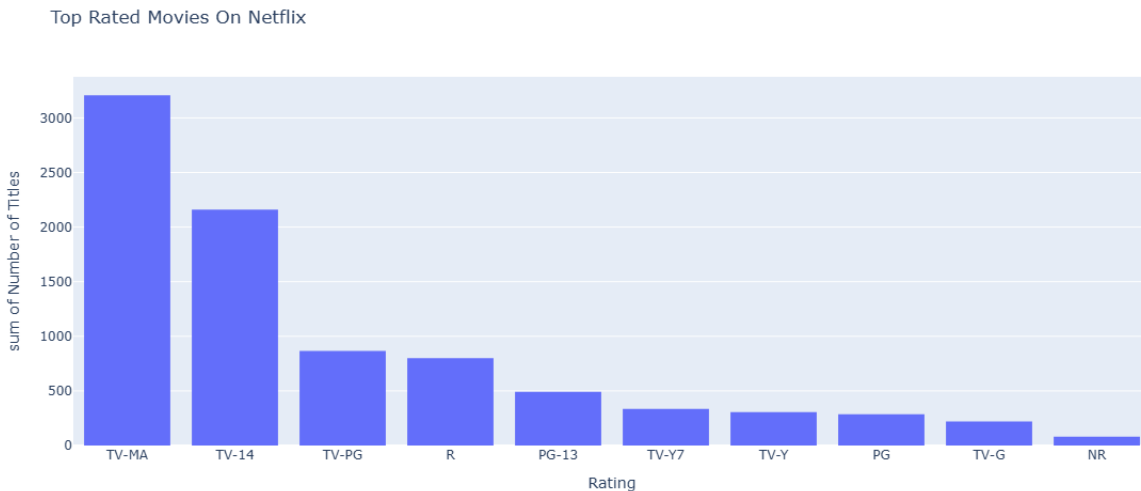
OUTCOME:-

- Earlier years (2009–2013) show relatively shorter average movie durations, indicating more compact content.
- From 2015 onwards, average durations begin to increase steadily.
- Post-2018, the average runtime consistently exceeds 100 minutes, highlighting a shift toward longer movies.
- The upward trend suggests changing viewer preferences and greater investment in feature-length, high-production movies.

Top Rated Movies On Netflix

TASK 10 :- IDENTIFY AND PRESENT TOP-RATED MOVIES OR TV SHOWS BASED ON USER RATINGS.

```
import plotly.express as px
px.histogram(g.head(10),x="rating",y="count",
             title="Top Rated Movies On Netflix ",
             labels={"count": "Number of Titles", "rating": "Rating"})
```



OVERVIEW:-

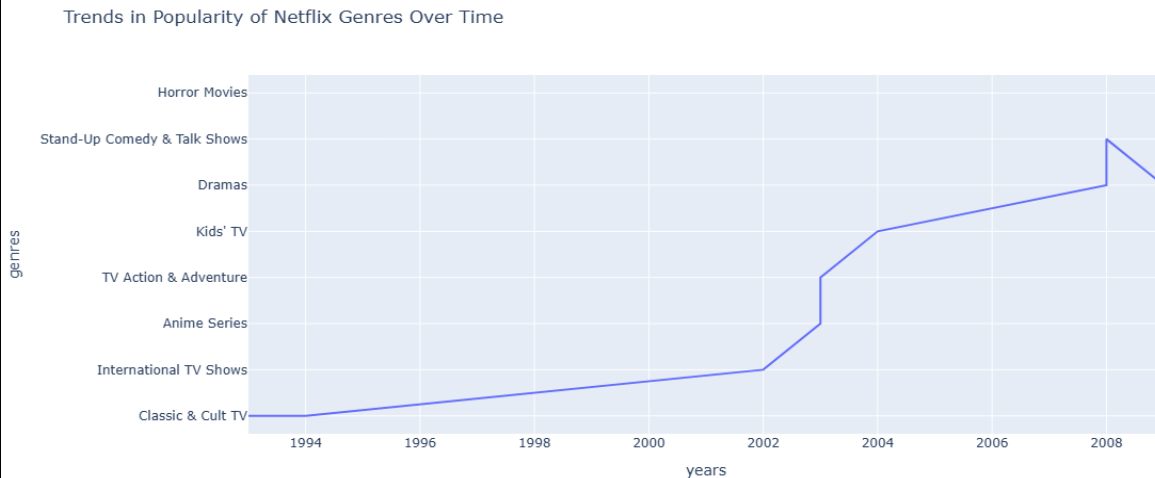
This Plotly bar chart displays the distribution of top-rated movies on Netflix by content rating. The x-axis represents rating categories (TV-MA, TV-14, PG-13, etc.), while the y-axis shows the number of highly rated titles within each rating group.

OUTCOME:-

- TV-MA dominates the chart, indicating that most top-rated movies are targeted at mature audiences.
- TV-14 follows closely, showing strong performance among teen-friendly content.
- Ratings such as TV-PG, R, and PG-13 have moderate representation.
- Kids-focused ratings (TV-Y, TV-Y7, TV-G) account for a smaller share of top-rated movies.
- NR (Not Rated) has minimal presence, suggesting limited popularity or classification.

Trends in Popularity of Netflix Genres Over Time

```
# TASK 11 :- ANALYZE TRENDS IN THE POPULARITY OF DIFFERENT GENRES OVER TIME.  
import plotly.express as px  
px.line(h.head(10),  
        x='year_added',  
        y='listed_in', labels = {"year_added": "years", "listed_in": "genres"},  
        title='Trends in Popularity of Netflix Genres Over Time')
```



OVERVIEW:-

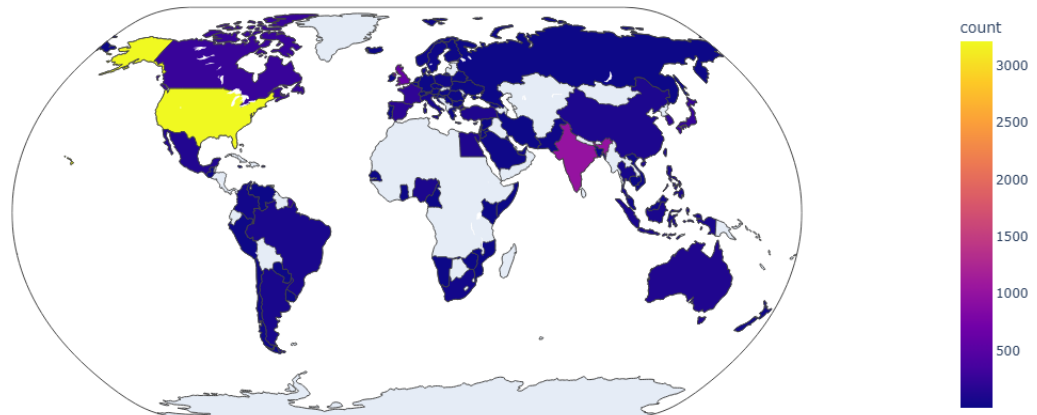
This Plotly line chart illustrates trends in the popularity of Netflix genres over time. The x-axis represents years, while the y-axis lists different genres, showing how genre popularity has evolved across different periods.

OUTCOME:-

- Classic & Cult TV shows early presence but grows slowly over time.
- International TV Shows show steady growth, reflecting Netflix's expansion into global markets.
- Anime Series and TV Action & Adventure gain popularity in later years, indicating rising demand for genre-specific content.
- Kids' TV and Dramas show strong upward trends, highlighting consistent viewer interest.
- Stand-Up Comedy & Talk Shows and Horror Movies emerge more prominently in recent years, suggesting diversification in content offerings.

DISTRIBUTION OF CONTENT ACROSS DIFFERENT COUNTRIES AND REGIONS.

```
# TASK 12 :- FURTHER EXPLORE THE DISTRIBUTION OF CONTENT ACROSS DIFFERENT COUNTRIES AND REGIONS.  
import plotly.express as px  
px.choropleth(i,locations= i["country"],locationmode="country names",  
              projection = "natural earth",color = i['count'])
```



OVERVIEW:-

This Plotly choropleth world map shows the global distribution of Netflix content by country. Each country is colored based on the number of titles available, with darker or brighter shades indicating higher content volume.

OUTCOME:-

- a) The United States stands out as the largest content contributor, dominating the Netflix library.
- b) India shows strong content representation, highlighting rapid growth in regional and local-language content.
- c) Europe (UK, France, Germany) and parts of Asia-Pacific contribute moderately, reflecting Netflix's international expansion.
- d) Many countries in Africa and smaller markets have lower content counts, indicating emerging or underrepresented regions.

POTENTIAL CORRELATIONS BETWEEN VARIABLES (E.G., RATINGS AND DURATION).

```
# TASK 13 :- INVESTIGATE POTENTIAL CORRELATIONS BETWEEN VARIABLES (E.G., RATINGS AND DURATION).
import plotly.express as px
px.box(movies,x='rating_numeric',y='duration_min',
       title='Distribution of Movie Duration Across Ratings',color = "rating_numeric")
```



OVERVIEW:-

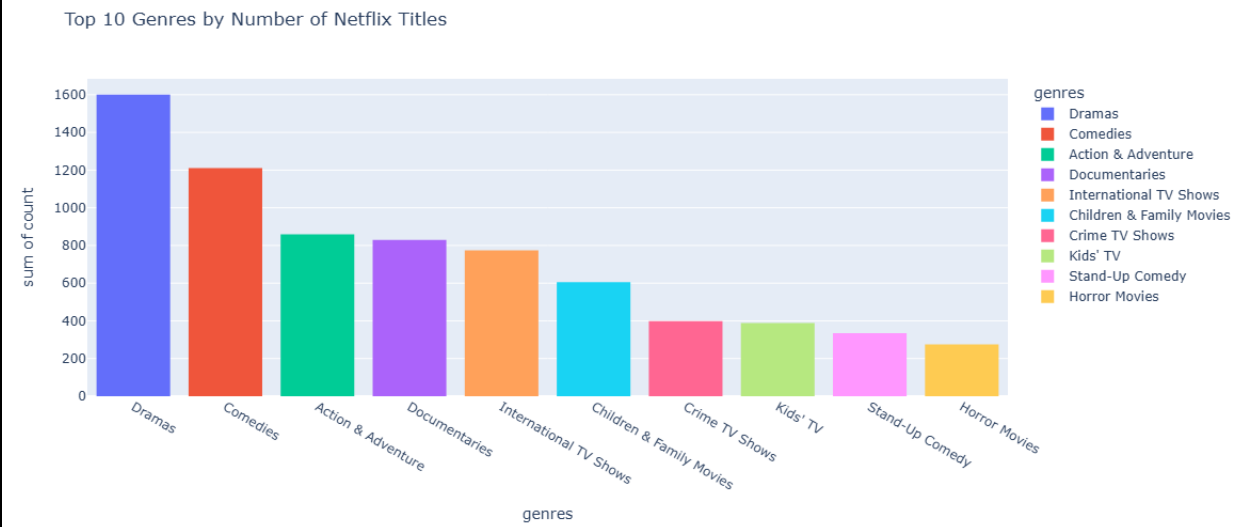
This Plotly box plot visualizes the distribution of movie durations across different rating categories (converted into numeric ratings). The x-axis represents rating levels, while the y-axis shows movie duration in minutes, highlighting central tendency, spread, and outliers for each rating group.

OUTCOME:-

- Lower rating categories tend to have shorter movie durations, indicating simpler or lighter content.
- Mid to higher ratings (3–5) show longer median durations, suggesting more detailed storytelling.
- Higher-rated movies exhibit greater variability in duration, including several long-duration outliers.
- Rating category 4 and 5 contain the longest movies, reflecting higher production depth and audience engagement.

(A) UNIQUE GENRES BY NUMBER OF NETFLIX TITLES

```
# TASK 14 :- EVALUATE THE DIVERSITY OF CONTENT BY ANALYZING THE NUMBER OF UNIQUE GENRES AND CATEGORIES.  
import plotly.express as px  
px.histogram(k.head(10),x='listed_in',y='count',color = "listed_in",  
             labels = {"listed_in": "genres"},  
             title='Top 10 unique Genres by Number of Netflix Titles')
```



OVERVIEW:-

This Plotly bar chart highlights the Top 10 Netflix genres by number of titles. The x-axis represents different genres, while the y-axis shows the total count of Netflix titles in each genre, making it easy to compare genre popularity.

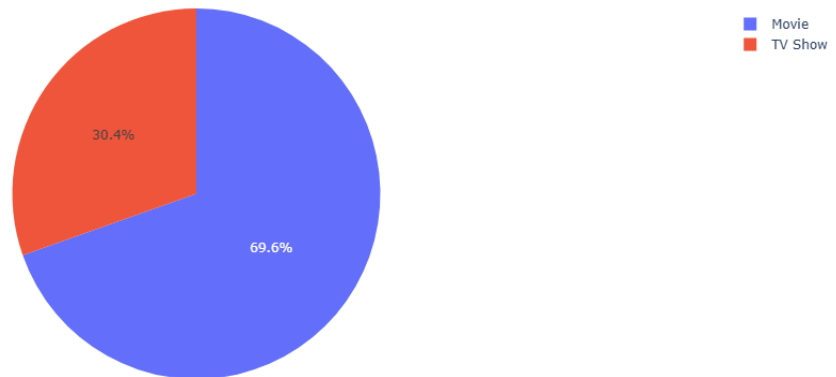
OUTCOME:-

- Dramas dominate the Netflix catalog, having the highest number of titles.
- Comedies and Action & Adventure follow, indicating strong demand for entertainment and high-energy content.
- Documentaries and International TV Shows also have substantial representation, reflecting audience interest in informative and global content.
- Children & Family Movies and Kids' TV show moderate counts, supporting family-oriented viewing.
- Stand-Up Comedy and Horror Movies have lower counts, suggesting niche but loyal audiences.

(B) DISTRIBUTION OF CONTENT CATEGORIES ON NETFLIX

```
# TASK 14 :- EVALUATE THE DIVERSITY OF CONTENT BY ANALYZING THE NUMBER OF UNIQUE GENRES AND CATEGORIES
import plotly.express as px
px.pie(category_counts, names='type', values='count',
        title='Distribution of Content Categories on Netflix')
```

Distribution of Content Categories on Netflix



OVERVIEW:-

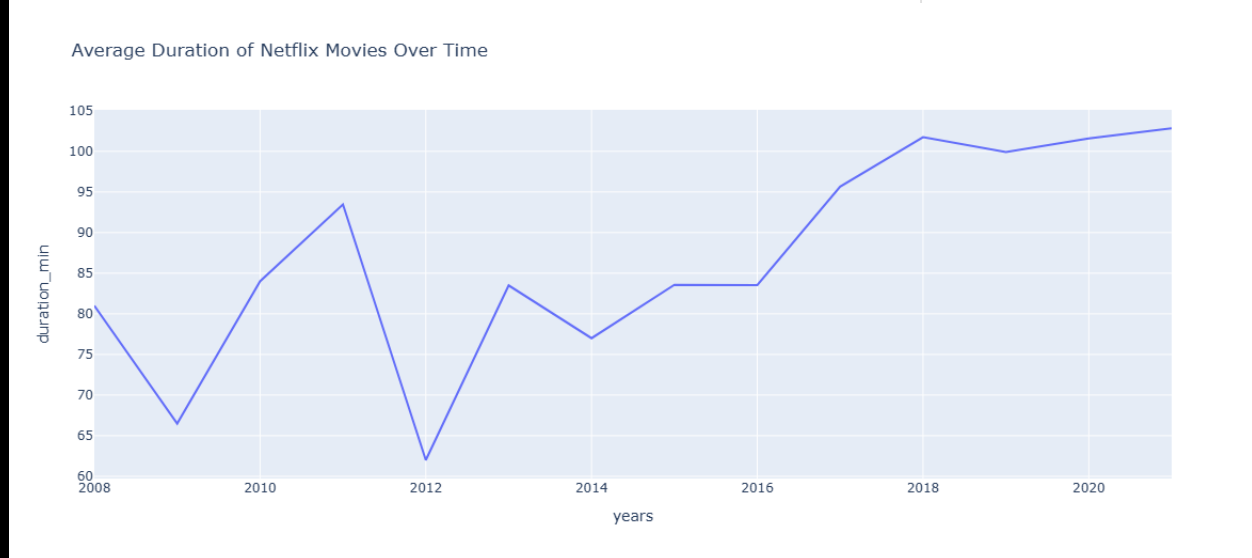
This Plotly pie chart represents the distribution of content categories on Netflix, comparing Movies and TV Shows. Each slice shows the proportion of total content belonging to each category.

OUTCOME:-

- a) Movies make up the majority of Netflix's catalog, accounting for around 70% of total content.
- b) TV Shows contribute roughly 30%, indicating a smaller but significant share.
- c) The clear dominance of movies highlights Netflix's strong focus on film-based content.

(A) AVERAGE DURATION OF NETFLIX MOVIES OVER TIME

```
# TASK 15 :- EXPLORE HOW THE CHARACTERISTICS OF CONTENT (E.G., DURATION, RATINGS) HAVE EVOLVED OVER THE YEARS.  
import plotly.express as px  
px.line(M,x='year_added',y='duration_min',  
        labels = {"year_added" : "years"},  
        title='Average Duration of Netflix Movies Over Time')
```



OVERVIEW:-

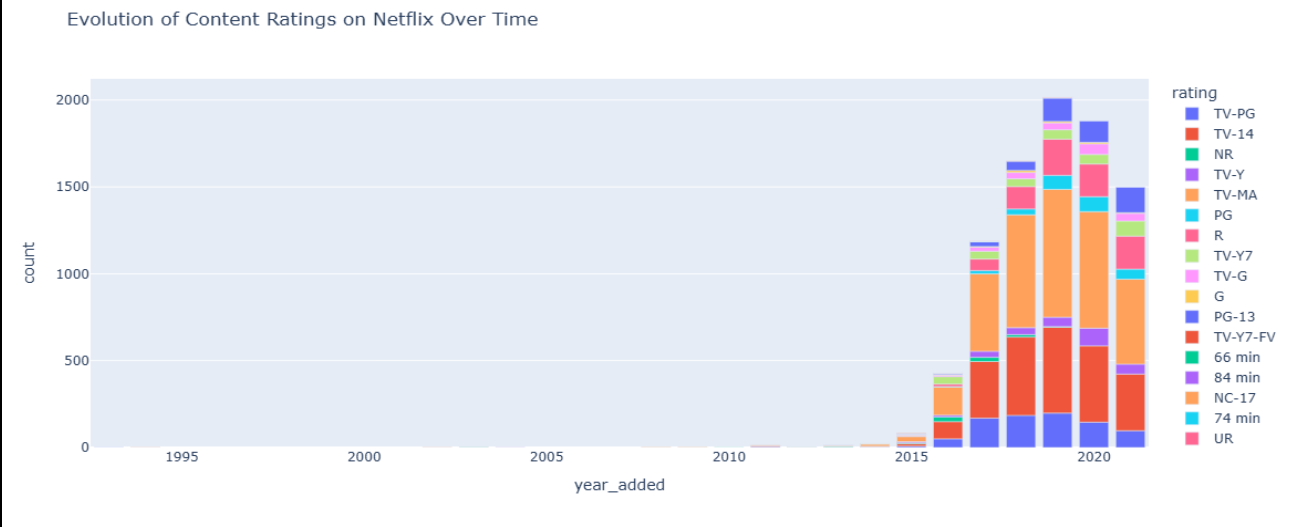
This Plotly line chart illustrates the average duration of Netflix movies over time. The x-axis represents years, while the y-axis shows the average movie duration (in minutes), helping track how movie length has evolved.

OUTCOME:-

- a) Average movie duration fluctuates in the early years (2008–2012), indicating experimentation with shorter formats.
- b) From 2013 onwards, movie duration shows a steady upward trend.
- c) Post-2017, average durations consistently exceed 95–100 minutes, reflecting a shift toward longer movies.
- d) Minor dips occur, but the overall pattern shows growth in runtime.

(B) Evolution of Content Ratings on Netflix Over Time

```
# # TASK 15 :- EXPLORE HOW THE CHARACTERISTICS OF CONTENT (E.G., DURATION, RATINGS) HAVE EVOLVED OVER THE YEARS.  
import plotly.express as px  
px.bar(rating_trend,  
       x='year_added',  
       y='count',  
       color='rating',  
       title='Evolution of Content Ratings on Netflix Over Time')
```



OVERVIEW:-

This Plotly stacked bar chart illustrates the evolution of Netflix content ratings over time. The x-axis represents the year content was added, while the y-axis shows the count of titles, segmented by different content ratings (TV-MA, TV-14, PG, etc.). It helps visualize how rating composition has changed year by year.

OUTCOME:-

- Content additions are minimal before 2015, followed by a sharp rise after 2016, reflecting Netflix's rapid expansion phase.
- TV-MA and TV-14 dominate in recent years, indicating a growing focus on mature and teen-oriented content.
- Family-friendly ratings (TV-Y, TV-Y7, TV-G) increase gradually but remain a smaller portion overall.
- Less common ratings (NR, UR, NC-17) contribute negligible volumes, showing limited emphasis.

conclusion

This exploratory data analysis of the Netflix dataset provides a comprehensive understanding of Netflix's content strategy, growth patterns, and audience targeting over time. The dataset, consisting of over 8,800 titles, was carefully cleaned by handling missing values in key fields such as director, cast, country, date added, rating, and duration, ensuring reliable and consistent analysis.

A) Content Composition & Categories

The analysis reveals that Movies dominate Netflix's catalog ($\approx 70\%$), while TV Shows account for around 30%, highlighting Netflix's stronger focus on standalone content while still maintaining a substantial episodic library. This balance allows Netflix to cater to both binge-watchers and casual viewers.

B) Genre Distribution Insights

Genre-wise exploration shows that Dramas and Comedies are the most prevalent genres, followed by Action & Adventure, Documentaries, and International TV Shows. This indicates Netflix's emphasis on emotionally driven storytelling and light entertainment, while also investing heavily in global and region-specific content, reflecting its international expansion strategy. Niche genres such as LGBTQ Movies, Sports Movies, and Sci-Fi TV content exist but form a very small fraction of the catalog.

C) Temporal & Growth Trends

The release year and date-added analysis highlight a sharp rise in content additions after 2015, peaking between 2017 and 2020, which aligns with Netflix's aggressive global growth and original content push. Monthly trends show that Netflix consistently adds content throughout the year, with noticeable spikes in certain months, suggesting strategic release planning to maximize viewer engagement.

D) Geographical Distribution

Geographical analysis confirms that the United States is the largest content producer, followed by India and the United Kingdom. The strong presence of Indian content reflects Netflix's growing focus on emerging markets and regional storytelling. Overall, the platform demonstrates wide global coverage, reinforcing its identity as a truly international streaming service.

E) Ratings & Audience Targeting

Content rating analysis indicates a clear skew toward mature audiences, with TV-MA and TV-14 being the most common ratings. This suggests Netflix prioritizes content aimed at young adults and adults, while still maintaining a reasonable catalog for children and family audiences through TV-Y and TV-G rated content.

F) Duration & Content Evolution

The analysis of movie duration over time reveals a gradual increase in average movie length, especially after 2016. This trend suggests a shift toward more in-depth storytelling and higher production value films. Additionally, the relationship between ratings and duration shows that mature-rated movies generally have longer runtimes, indicating more complex narratives.

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Thank you

<https://colab.research.google.com/drive/1TgisQsWdijfgFhkupLUySGgJMzKkaHQy?usp=sharing>